UNIT 1 · TOPIC 4(B)

Inter-Cultivation Equipment

Syllabus topic: Equipment for fertilizer application and inter-cultivation

Objectives

Weed control (the biggest benefit — Kepner ch. 12), soil mulch, aeration and earthing-up.

Equipment

Group	Equipment
Hand tools	Khurpi, hand hoe, wheel hoe (single/double wheel with interchangeable sweeps/tines; capacity ~0.015–0.02 ha/h), peg/spike weeder.
Paddy	Cono-weeder — works between rows in a puddled field; two conical rotors with smooth and serrated blades.
Dryland	Dry-land weeders, blade hoe, sweep cultivator — V-sweeps cut weeds below the surface with the least soil disturbance and moisture loss.
Power weeders	Engine-operated rotary weeders (1–2 hp, petrol/kerosene) for row crops; tractor-mounted row-crop cultivators with shields to protect young plants.
Rotary hoe	High-speed blind cultivation before/after crop emergence (Kepner ch. 12).

Good practice

- Cultivate shallow (**3–5 cm**) to avoid pruning roots.
- Flame cultivation and thinning (Kepner ch. 12): use selective flaming once cotton stems have hardened.

Practice MCQs

Q35. Cono-weeder is used in:

(a) Dryland wheat
(b) Puddled paddy between rows
(c) Sugarcane earthing
(d) Cotton topping

Q36. Sweep-type cultivators are favoured in dryland inter-culture because they:

(a) Invert soil
(b) Cut weeds below surface with least moisture loss
(c) Work deep
(d) Form ridges

Answers

Q35 — (b) Puddled paddy between rows

Q36 — (b) Cut weeds below surface with least moisture loss